Reva Shankar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.23
Banyan International School, Jammu	Class XII (CBSE)	2019	61.40%
Banyan International School, Jammu	Class X (CBSE)	2017	85.50%

M.TECH THESIS

Experimental Study on 316L Stainless Steel Produced by Laser Powder Bed Fusion

- Investigating the influence of **Laser Powder Bed Fusion processing parameters and build direction** on the microstructure and mechanical properties of 316L stainless steel.
- Characterizing **as-printed 316L stainless steel** through powder and phase analysis, XRD, SEM, microstructure, tensile, hardness, and wear testing to establish baseline properties.
- Investigating the effects of **nitriding and carburizing** on separate 316L samples through post-treatment characterization and comparison of microstructure, hardness, mechanical, and wear properties.

PROJECTS

CFD Analysis and Performance Comparison of Square and Cylindrical Pin-Fin Arrays

- Developed symmetric computational models of square and cylindrical pin-fin arrays using **ANSYS SpaceClaim**, followed by mesh generation and quality assessment.
- Performed steady-state **conjugate heat-transfer simulations** in **ANSYS Fluent** using the **k- SST turbulence model** to analyze flow and thermal fields under varying inlet velocities.
- Compared square and cylindrical fin configurations through velocity/temperature contours, **heat-transfer coefficient, bypass factor and fin-efficiency** analysis, demonstrating superior thermal performance of cylindrical fins.

Piston-Cylinder-Crank Mechanism | Design & Additive Manufacturing

- Designed a functional **piston-cylinder-crank mechanism in SolidWorks**, incorporating dimensional accuracy, assembly constraints and motion verification.
- Fabricated the mechanism using **FDM 3D printing** and assembled the components to evaluate fit, motion and mechanical integrity.
- Applied a **Design for Additive Manufacturing (DfAM)** workflow from CAD modelling to fabrication and functional validation.

Performance Evaluation of Solar Powered Water Temperature Controlling System for Hydroponics Using a Thermoelectric Module

- This project explores a solar-powered thermoelectric system for precise temperature control in hydroponic farming.

RESEARCH PUBLICATION

Review of Recent Advancements in Thermoelectric Technologies

- Reviewed recent advancements in **thermoelectric cooling technologies**, with emphasis on Peltier modules and their potential as alternatives to conventional refrigeration systems.

INTERNSHIP

IIT Jammu | RISE-UP Internship Program

- Designed and executed a project focused on the **development of an oxygen concentrator**.
- Gained hands-on experience in **component and system integration** for efficient oxygen concentration, delivery and safety.

CIIIT (TATA Technologies), Jammu | Summer Intern

- Gained hands-on exposure to **Electric Vehicles and Connected Autonomous Vehicles**, including motors, DC-DC converters, choppers and DC drives.
- Worked on **e-rickshaw assembly** and studied its major electrical and mechanical components.
- Performed battery performance analysis with emphasis on **State of Charge (SOC)** for electric-vehicle applications.

SKILLS

- **Technical Skills:** SolidWorks, Additive Manufacturing, Laser Powder Bed Fusion (LPBF), CFD, ANSYS Fluent, ANSYS SpaceClaim, FDM 3D Printing
- **Programming:** C, C++, Python
- **Industrial Automation:** PLC Programming, HMI, SCADA
- **Core Engineering:** Mechanical Design, Manufacturing Processes, Problem Solving, Team Leadership

POSITIONS OF RESPONSIBILITY

- **Class Representative, B.Tech:** Coordinated communication between students and faculty and supported class-level academic coordination.
- **AET Club Coordinator:** Coordinated club activities and supported the planning and execution of student activities and events.
- **Atelier Photography Club:** Volunteered in photography-related activities and events.
- **Art Club Volunteer:** Assisted in organizing club activities and supporting creative events and initiatives.

EXTRACURRICULAR

- **Sports & Fitness:** Basketball, Volleyball, Swimming, Wushu, Regular Gym Training
- **Hobbies:** Cooking, Photography, Sketching, Keyboard Playing, e-Sports